Calendar
REAL VARIABLES I
Fall 2026

I will put all deadlines on your Blackboard Calendar. Please check it regularly!

0.1 Key university dates

For official University dates, see the Fall 2026 academic calendar and the Registrar's final-exam schedule.

Date	Event
Monday, August 17	First day of classes
Friday, August 21	Last day to add this course through the standard add period
Friday, August 28	Last day to drop without a W
Thursday, December 3	Last day of classes and last day to withdraw from all classes

0.2 Exams

Date	Event
Week of Wednesday, October 12 - Oct 18, 2026, TENTATIVE	Written midterm (two hours, outside t
Wednesday, December 9, 2026, 10:15 - 12:15	Final-exam period

0.3 Daily lecture log

	Wk	Lec
		L1: Riemann integration and its limitations
2026-08-17	1	Review: the Riemann integral; theorem+proof: continuous functions are Riemann integ
		L2: Countability and unbounded functions
2026-08-19		Proposition + proof: continuous functions with L1 norm are not a complete metric spa
		L3: Countability, the extended reals, and sums of non-negative numbers
2026-08-21		Review: fat rationals; sketch: Riemann integrable functions with respect to L^1 are inco
		L4: Outer measures TENTATIVE
2026-08-24	2	Review: continuity of arbitrary non-negative sums with respect to increasing index sets
		L5: Examples and Lebesgue outer measure on $\mathbb{R}^d$ TENTATIVE
2026-08-26		Defn: Lebesgue outer measure on $\mathbb{R}^d$ Compute outer measures generated by singleton c
		L6: σ algebras and measures TENTATIVE
2026-08-28		Defn: σ additivity The Vitali counterexample. Motivations for measures. Defn: σ algeb
		L7: The volume of a rectangle TENTATIVE
2026-08-31	3	Prove that the Lebesgue outer measure of a rectangle is its volume, using compactness
		L8: The target: countable additivity TENTATIVE
2026-09-02		Contrast finite additivity, countable additivity, countable subadditivity, and countable s
		L9: The Vitali obstruction TENTATIVE
2026-09-04		Construct a Vitali set in $[0, 1]$ using equivalence modulo rational translation. Use transl
		No class: No class: Labor Day TENTATIVE
2026-09-07	4	*Labor Day holiday; no class.*
		L10: Algebras, σ-algebras, and generators TENTATIVE
2026-09-09		Define rings, algebras, and σ-algebras and prove their standard closure properties. Wor
		L11: Measures and their basic inequalities TENTATIVE
2026-09-11		Define measures on σ-algebras. Derive finite additivity, monotonicity, subtraction when
		L12: Continuity of measures and Borel sets TENTATIVE
2026-09-14	5	Prove continuity from below for arbitrary measures and continuity from above under th
		L13: Carathéodory measurability I TENTATIVE
2026-09-16		Motivate the splitting criterion from the missing superadditive inequality. Define Carat
		L14: Carathéodory measurability II TENTATIVE
2026-09-18		Follow a fixed proof roadmap: first obtain the finite disjoint-union identity, then iterate
		L15: σ-finite measures and Dynkin's π–λ theorem TENTATIVE
2026-09-21	6	Define finite and σ-finite measures, π-systems, and Dynkin (λ-)systems. Prove Dynkin's
		L16: Uniqueness of measures TENTATIVE
2026-09-23		Prove the uniqueness criterion for two finite measures agreeing on a generating π-system
		L17: Measurable functions TENTATIVE
2026-09-25		Begin the next unit. Define measurable spaces and measurable functions, including Bor
		L18: Operations and limits of measurable functions TENTATIVE
2026-09-28	7	Characterize real-valued measurability using level sets. Prove closure under algebraic op
		L19: Simple-function approximation TENTATIVE
2026-09-30		Define indicator and simple measurable functions. Construct an increasing sequence of
		L20: Distribution functions and layer cake TENTATIVE

TENTATIVE: Future lecture topics and their placement may change as the course develops.